

LESSON 05

Arduino Fundamentals

<https://youtu.be/JnJIKX5J0Cc?si=okGV43s1KrbdgXzN>

<https://youtu.be/SX8z3-BEuWQ?si=ogRrtr8yZl-ZFKWY>

<https://youtu.be/ynAvySNCi-0?si=Mmpxcu9Vex71KvhW>

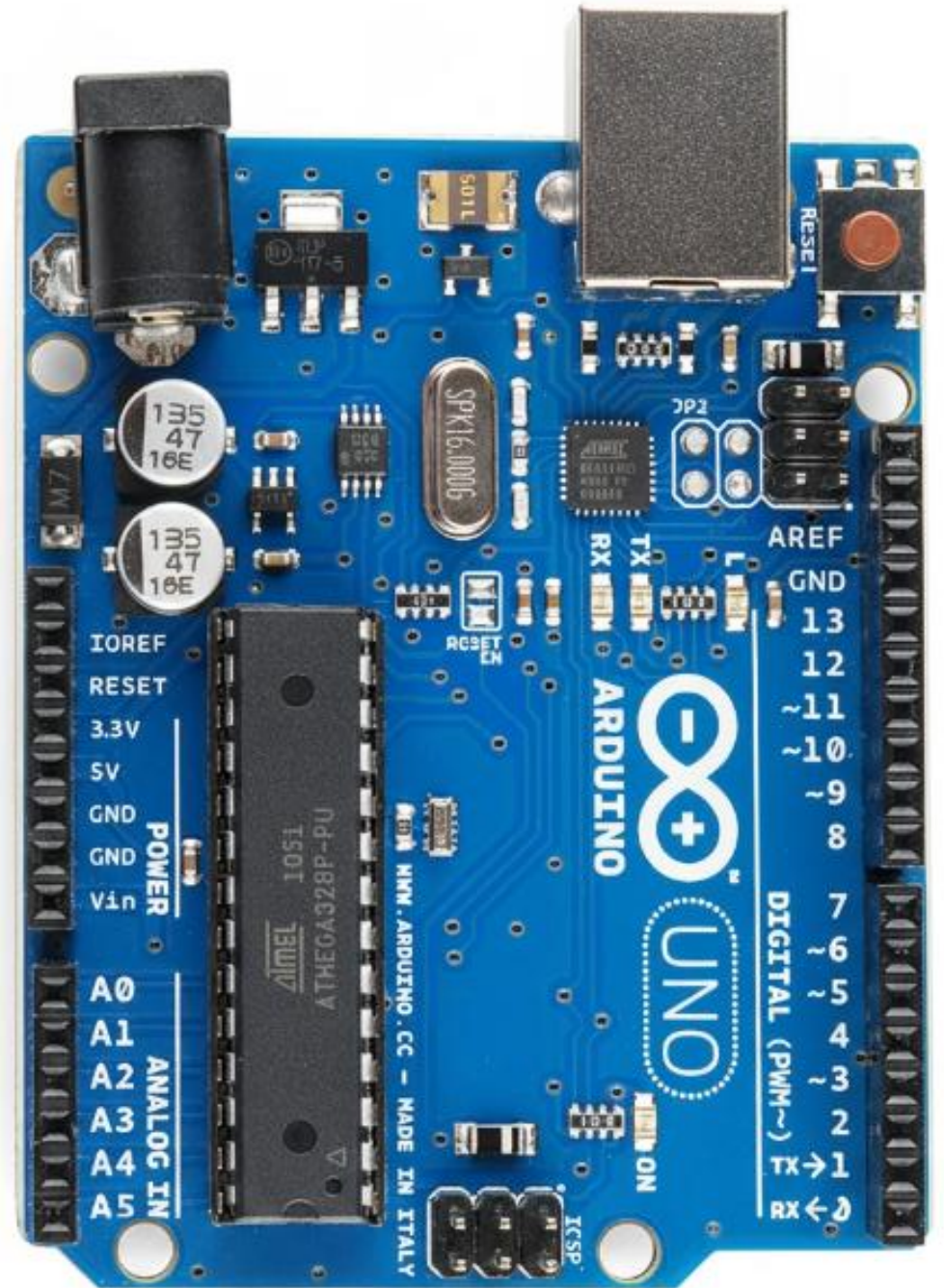
<https://youtu.be/EVDnYnAyktw?si=GNGQ4dFUMsDcCKM6>

<https://youtu.be/m8ov5S78AIY?si=TPGUTO5QZdKriB-2>

<https://youtu.be/Q5A2vsa2B-g?si=Ml0D96XMUI2zVZ2E>

https://youtu.be/oOBxocRbdbo?si=M_kBWzziXQCrFWX8

From pins and power to your first blinking LED.



THE BIG IDEA

Arduino connects code to the real world.

A small computer for physical projects: it reads what is happening, makes a decision, and controls what happens next.

Board + IDE + sketch work together.



Reads inputs

Buttons, sensors, and switches tell the board what is happening.



Runs a sketch

A short program makes decisions in code.



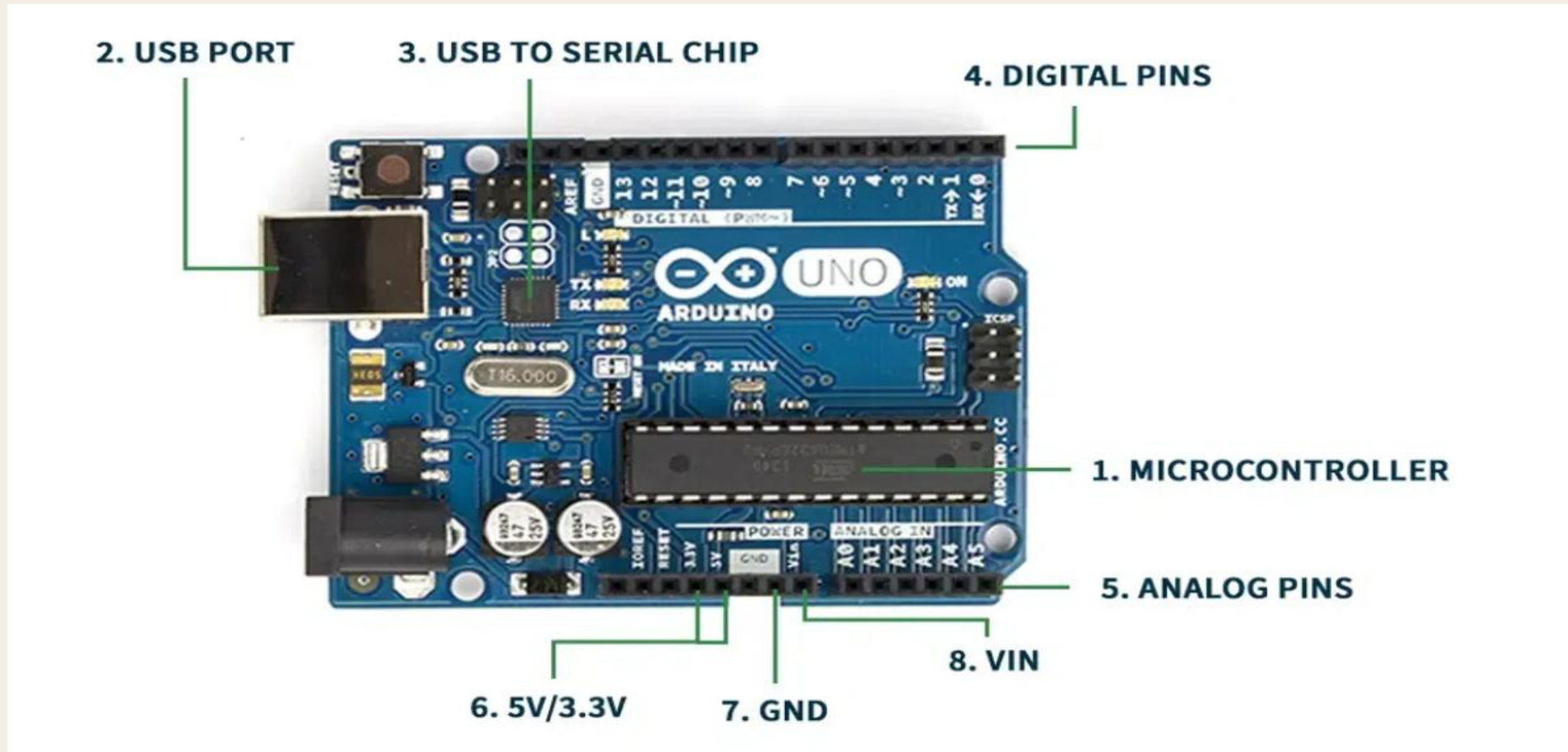
Controls outputs

LEDs, buzzers, motors, and displays respond.

MEET THE BOARD

Know the UNO before you wire it.

Find the places where code enters, signals travel, measurements happen, and power is shared.



FROM IDEA TO BOARD

The Arduino IDE is the path from idea to board.

Choose the correct board and port before uploading. The IDE turns your sketch into instructions the board can run.

1

Write

Create a sketch in the editor.

2

Compile

Check for errors and build instructions.
error check

3

Upload

Send the compiled sketch through USB to the Arduino.

4

Run

The board starts the new sketch.

TWO WAYS TO READ

Digital asks a yes-or-no question; analog measures a range.

DIGITAL

LOW or HIGH

Two clear states. Use `digitalRead()` to read and `digitalWrite()` to control.

Switches and LEDs fit this yes-or-no model.



open / closed

ANALOG

A changing value

Measure a range from a sensor or potentiometer with `analogRead()`.

Brightness, position, and light level can vary.



less / more

Every Arduino sketch has a setup and a loop.

One opening step prepares the board. Then one repeating step keeps the project alive.



setup()

Runs once when the board starts or resets.



loop()

Runs repeatedly while the board has power.



Read

Check inputs and signals.



Update

Make decisions and change outputs.

PIN BEHAVIOR

`pinMode()` tells a pin how to behave.

Choose whether the pin listens, drives, or listens with an internal pull-up resistor.

01

INPUT

The pin listens for an external signal, such as a switch.

02

OUTPUT

The pin drives a component, such as an LED.

03

INPUT_PULLUP

The pin listens with the Arduino's internal pull-up resistor.

Open = typically HIGH ·
connected to GND = LOW

```
pinMode(7, INPUT_PULLUP);
```

CODE SHAPE

Code punctuation gives the sketch its shape.

01

//

Starts a single-line comment.

```
// read the switch
```

02

/* */

Wraps a multi-line comment.

```
/* setup notes */
```

03

()

Holds a function's inputs or conditions.

```
delay(1000);
```

04

{ }

Marks the beginning and end of a block.

```
void loop() { }
```

Comments explain intent; brackets show which instructions belong together.

FIRST PROJECT

Make the built-in LED blink.

Use pin 13, set it as an output, then alternate HIGH and LOW inside `loop()`.

A working sketch is a conversation between code and light.

```
const int ledPin = 13;
```

```
void setup() {  
    pinMode(ledPin, OUTPUT);  
}
```

```
void loop() {  
    digitalWrite(ledPin, HIGH);  
    delay(1000);  
    digitalWrite(ledPin, LOW);  
    delay(1000);  
}
```

setup() prepares the pin once.

loop() repeats.

HIGH / LOW turns the LED on/off.

INSTALLING THE ARDUINO IDE

Step 1: Download the Software

1. Go to the official website: arduino.cc/en/software.
2. Select the installer for your operating system (**Windows**, **macOS**, or **Linux**).
3. Download the latest version of the **Arduino IDE 2.x**.

Step 2: Install the IDE

1. Run the downloaded installer file.
 2. Follow the setup wizard prompts and accept the license agreement.
- Important:** Ensure driver installation options (USB drivers) are checked during setup. Complete the installation and open the **Arduino IDE**.

Step 3: Connect and Select Your Board

1. Plug your **Arduino UNO** into your computer using a **USB Cable**.
2. In the top toolbar of the IDE, click the **Board Dropdown**.
3. Select **Arduino Uno** from the list.
4. Select the corresponding **COM Port** (Windows) or **/dev/cu.usbmodem** (macOS).

TRY · OBSERVE · CHANGE

Upload, observe, and debug one change at a time.

- ☐ **Select** Arduino UNO and the correct USB port.
- ☐ **Compile** before uploading so errors are easier to locate.
- ☐ **Observe** the LED and compare the result with the sketch.
- ☐ **Check** pin number, pinMode(), polarity, USB connection, and brackets.
- ☐ **Change one thing**, upload again, and observe what changed.

Small experiments turn code into understanding.